

Using the Plugin

Opening the dialog

Click the plugin toolbar icon or navigate to Plugins → EGMS Time Series Concatenation. The dialog will open and automatically populate the layer drop-down lists with all vector layers currently loaded in your QGIS project.

Selecting inputs

Period 1 CSV — Select the first observation period. Use the "Choose input layer" drop-down if the file is already loaded in QGIS or click **Browse** to pick a CSV from disk.

Period 2 CSV — Select the second observation period in the same way.

Clipping to an area of interest (optional)

Tick the Clip to AOI (optional) checkbox to enable spatial clipping. The AOI widget will be activated. Select a polygon Shapefile using the layer combo or the Browse button. The plugin will retain only MPs that fall within the AOI boundary before performing the concatenation.

Setting the output path

Click **Browse** next to "Output csv file" and choose a destination folder and filename. The file must have a **.csv** extension.

Running the process

Click **Run**. The dialog closes immediately and processing continues in the background via the QGIS task manager. Progress is shown in the QGIS status bar. When finished, the output layer is loaded automatically into the map canvas, and a success notification appears in the message bar.

Note: Processing time depends on the number of points and your hardware. For full EGMS tiles with millions of points, allow approximately one hour. You can continue using QGIS while the task runs.

Output

The output is a single CSV file and a corresponding QGIS point layer (EPSG:3035). Each row represents one persistent scatterer. The table below describes the output fields:

Field	Description
<code>pid</code>	Measurement point identifier (shared across periods).
<code>easting / northing</code>	Coordinates of the point in ETRS89-LAEA (EPSG:3035).
<code>mean_velocity_mmyr</code>	Linear velocity (mm/yr) derived from a least-squares fit over the full concatenated time series.

<code>offset_mm</code>	Systematic displacement offset (mm) between Period 1 and Period 2, estimated from the overlap window.
<code>YYYYMMDD</code> columns	Displacement values (mm) at each acquisition date, referenced to the first valid observation of each measurement point.

Note: All displacement values are referenced to zero at the first valid acquisition of each measurement point. The `mean_velocity_mmyr` field is computed by ordinary least-squares linear regression over all valid epochs.

Fusion method

The plugin estimates a systematic displacement offset for each measurement point using the acquisition dates common to both periods:

$$\text{offset} = \text{median}(\text{Period 2 values} - \text{Period 1 values}) \text{ over overlap dates}$$

Period 1 values are shifted by this offset before concatenation, so the two series meet smoothly. If no overlap exists, no offset correction is applied. The output displacement series is then re-referenced so that the first valid value of each point equals zero.

Troubleshooting

Symptom	Likely cause / action
No points in output	No common pid values between the two CSVs, or AOI does not overlap any points. Verify that both files cover the same EGMS tile.
Output layer not loaded	QGIS may have been unable to parse the CSV. Check the QGIS Log Messages panel (View → Panels) for details.
Task fails immediately	A required Python package (pandas, geopandas, numpy) may be missing. Install it via OSGeo4W Shell or the QGIS Python console.
Fewer points than expected	Points with mismatched pixel/line coordinates are automatically removed. This is expected behaviour when grid alignment differs between periods.
Missing inputs warning	Period 1 CSV, Period 2 CSV, or the output path is empty. All three are mandatory.